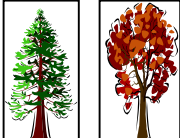


Land Use and Land Cover of Canada

The landmass of Canada is made up of a wide variety of land covers, from bare rock to dense forest and everything in between. Land cover, as its name implies, is the material that covers the surface of the earth. Some land covers are rich in biological materials such as grasses, trees and crops. Others consist of only lichen, mosses, bare rock or ice and snow. With this great diversity of land cover there also comes a wide range of possible uses. First, let's look at the 31 land cover classes adapted by Canadian scientists in the Land cover of Canada Classification using the AVHRR (Advanced Very High Resolution Radiometer) mosaic of Canada.

MAP LEGEND FOR LAND COVER OF CANADA 1995*

Group	Sub-group	Class	Description
Forest Land			Land dominated by trees exceeding 5 m with crown density (percentage of the surface covered by projected tree crown perimeters) greater than 10%.
Evergreen Needleleaf Forest			Land occupied by forest containing more than 80% needleleaf trees
	High Density	High Density (1)	Evergreen needleleaf forest (southern boreal) with crown density of the needleleaf species above 70%.
	Medium Density		Evergreen needleleaf forest with crown density of the needleleaf species between approximately 40-70%.
		Medium Density Southern Forest (2)	Often occurs within, or adjacent to, high density forest. It has a higher proportion of broadleaf trees or shrubs (woody plants less than 2-3 m high) than high density forest
		Medium Density Northern Forest (3)	Shrubs and lichens commonly present in the understory.
	Low Density		Evergreen forest with crown density of the needleleaf species approximately 10-40%.
		Low Density Southern Forest (4)	Higher proportion of broadleaf trees or shrubs species than high density forest
		Low Density Northern Forest (5)	Shrubs and lichens commonly present in the understory
Deciduous Broadleaf Forest		Deciduous Broadleaf Forest (6)	Concentrated occurrence, generally with a high crown density, primarily the shade-tolerant hardwood species (maples, yellow birch).
Mixed Forest			Mixed forest with the proportion of evergreen needleleaf trees exceeding approximately 60%. Land occupied by forest containing 20-80% evergreen needleleaf or deciduous broadleaf trees
	Mixed Needleleaf Forest	Mixed Needleleaf Forest (7)	Mixed forest with the proportion of evergreen needleleaf trees exceeding approximately 60%
	Mixed Intermediate Forest		Mixed forest with the proportion of evergreen needleleaf (or deciduous broadleaf) trees approximately 40-60%

		Mixed Intermediate Uniform Forest (8)	Relatively uniform distribution of trees in the landscape, typically with a higher crown density.
		Mixed Intermediate Heterogeneous Forest (9)	Lower crown density or forest with a patchy distribution of trees in the landscape, typically after disturbance (due to natural or human intervention)
	<i>Mixed Broadleaf Forest</i>	Mixed Broadleaf Forest (10)	The proportion of deciduous broadleaf trees exceeding approximately 60%
Burns			Land previously occupied by forest, which was subject to fire. At present it may contain broadleaf or needleleaf trees with a tree crown density of less than 10% or standing dead trees.
		Recent Burns (11)	Burns with small amounts of green vegetation present, probably burned within the last 5 years
		Older Burns (12)	Burns with greater amount of green vegetation present, implying earlier fires.
Open Land			Land with a tree crown density of less than 10%.
Transition Treed Shrubland		Transition Treed Shrubland (13)	Land covered with woody shrubs typically appearing as dense cover of low broadleaf vegetation. May include tree species in early regeneration stages. Also includes shrubs in wetlands.
Wetland / Shrubland			
		High Density (14)	Shrubland with more than 60% cover density.
		Medium Density (15)	A mixture of shrubs (approximately 40-60%) and herbaceous cover. Some wetlands, especially fens.
Grassland		Grassland (16)	Land with herbaceous (non-woody) vegetation cover; tree or shrub cover is less than 10%. This class is limited to the prairie region.
Barren Land			Land containing less than 10% of trees. It often contains low shrubs, lichen, herbaceous vegetation cover, bare soil, rock, or small water bodies. It is found mostly north of the tree-line, but also after disturbance in southern areas, and in mountainous regions.
	<i>Shrub/Lichen-Dominated</i>		Barren land in which low shrubs and lichen are the dominant cover type.
		Lichen & Others (17)	
		Shrub/Lichen Dominated (18)	Shrub-dominated barren land. In northern boreal forest, it may represent low density needleleaf forest with lichen understory. North of the tree-line, this class may also include abundant water bodies.
	<i>Treeless</i>		Barren land north of the tree-line.
		Heather and Herbs (19)	Treeless barren land in which lichen is the prevalent vegetation cover. The landscape typically consists of a pattern of lichen, shrubs, bare soil, and rock outcrops.
		Low Vegetation Cover (20)	Treeless barren land in which vegetation cover (shrubs, lichen herbaceous) does not exceed approximately 40% of the ground cover.
		Very Low Vegetation Cover (21)	Treeless barren land in which vegetation cover (shrubs, lichen, herbaceous) does not exceed approximately 10% of the ground cover area.
		Bare Soil and Rock (22)	Treeless barren land with patches of permanent snow cover.

Developed Land			
Cropland			Land covered with herbaceous (typically annual) crops, which may contain a small proportion (less than 10%) of trees or shrubs.
		High Biomass (23)	Cropland dominated by crops with higher biomass, due to cover type (e.g., corn) or climate (adequate precipitation). May contain small proportions of other vegetation types (less than 10%).
		Medium Biomass (24)	Cropland dominated by crops with medium biomass, due to cover type or climate (sub-humid). Occurs mainly in the prairie region.
		Low Biomass (25)	Cropland dominated by crops with lower biomass, due to cover type (e.g., grain) or climate (semiarid region). Occurs mainly in the prairie region.
Mosaic Land			Land containing a mix of cropland, forest, shrubland, or grassland in which no one component comprises more than about 60% (by area) of the landscape.
		Cropland-Woodland (26)	Mosaic land in which cropland is more prevalent than forest cover (mostly broadleaf deciduous forest).
		Woodland-Cropland (27)	Mosaic land in which woodland is more prevalent than crop cover
		Cropland-Other (28)	Mosaic land in which cropland is more prevalent than other cover types. These could be forest, shrubland, or built-up areas. Lower green biomass than class 25.
Urban and Built-up		Urban and Built-up (29)	Land covered by buildings and other man-made structures.
Non-Vegetated Land			Land covered with water (in solid or liquid form).
Water		Water (30)	Land covered with liquid water
Snow/ ice		Snow/ ice (31)	Land covered with permanent ice or snow.

Each of these land covers can be associated with a land use. It is obvious that the forest region would be linked to forestry. However, there are both productive and non-productive forests. Productive forests are those, which can be harvested for lumber or wood pulp. Non-productive forests are those in which the trees are not large enough or the forest not dense enough to make harvesting profitable. Areas of recent forest fire burns are usually unproductive as well. Other classes that are self-explanatory are urban and built-up, non-vegetated land of water and snow and cropland. Many of the remaining classes are combinations in different proportions of woodland, cropland and transition forest.

Man has learned that land cover can be changed and modified to his own use. Forests can be cleared for agriculture or grazing. Almost any land can be converted to urban and built-up. But are these the wisest uses of our land resources? Will irreparable damage be done to our environment? Are the uses we give areas of land compatible with each other (e.g. a quarry or agricultural land next to a residential area)? Land planners face these issues regularly. Some of the basic land uses are listed below:

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|---------------------------|------------------------|
| 1. Built-up and urban | 2. Productive woodland |
| 3. Un-productive woodland | 4. Outdoor recreation |
| 5. Mines and quarries | 6. Cropland |
| 7. Orchards and vineyards | 8. Improved pasture |
| 9. Unimproved pasture | 10. Swamp, marsh |
| 11. Water | |



From this list you can see that some land, such as urban/built-up and cropland, have commercial value while others, such as swamp and marshes, while having no commercial value, are important to the environment and wildlife. In order to have an environmentally healthy land, we must learn to balance our land uses. Biological diversity is important to a sustainable environment. In order to achieve this, we must find a feasible way of living with nature and not overtaking it. By using land for urban or industrial development that is not suitable for forestry or cropland, we protect our economic and food resources. By preserving wetlands we ensure our watersheds are not disturbed and wildlife habitat is not destroyed. A land use plan that is well thought through is our best defense against a destroyed ecosystem.